F78PA Probability & Statistics A

Tutorial 10

1. A coin is weighted so that its probability of landing on heads is 20%. Suppose the coin is flipped 20 times.

   (a) Using Markov's inequality, find a bound for the probability it lands on heads at least 16 times.

   (b) Using Chebyshev's inequality, find the probability that at least 16 flips result in heads.

   (c) Compare (a) and (b) with the actual probability. Comment on the results obtained.

2. The average height of a raccoon is 10 inches and the standard deviation is 2 inches.

   (a) Using Chebyshev's inequality, find a lower bound for the probability that a raccoon is between 5 and 15 inches tall.

   (b) Now assume this distribution is normal. Determine the probability from part (a) using the normality assumption. How close was your bound to the true probability?

3. A student has entered a sack race and must jump 50 metres in a sack. Suppose that the length of each jump is random and exponentially distributed with mean equal to 1 metre and suppose that jump lengths are independent. Use the Central Limit Theorem to approximate the probability that the student must jump at least 60 times in order to finish the race.

4. Anna has a backlog of 100 emails to answer. Based on past experience, she estimates that the probability that it will take her more than one minute to answer any one email is 0.6. Use the Normal approximation to the Binomial distribution to approximate the probability that at least 65 of the 100 emails will take Anna more than one minute to answer.

5. A lecturer must mark 100 exam scripts. The lecturer estimates that the time that it takes her to mark each script is a random variable with mean 0.3 hours and variance 0.05 hours^2.

   (a) If the lecturer starts with a stack of 27 scripts, approximate the probability that the lecturer will finish marking the 27 scripts within 8 hours.

   Be sure to state the assumptions that you make in order to justify your calculation.

   (b) Determine the number of hours that the lecturer should allocate for marking the 100 scripts so that the probability that she will complete the marking within the allocated time is at least 0.95.

6. Suppose that X is a random variable with density $f_X(x) = 3x^2$ on the interval $(0, 1)$ and suppose that Y is a random variable with conditional density $f_{Y|X}(y|x) = \frac{1}{x}$ when $0 < y < x$ and $0 < x < 1$.

   (a) Describe the conditional distribution of Y given $X = x$ and *without calculation*, determine $\mathbb{E}(Y \mid X = x)$.

   (b) Use the Law of Total Expectation to determine $\mathbb{E}(Y)$.

   (c) Use the Law of Total Probability (for densities) to determine $F_Y(y)$, the distribution function for Y and derive the marginal density $f_Y(y)$ for Y from $F_Y(y)$.

7. Sam likes to use a certain type of ball point pen to take notes and write his tutorial solutions up. Suppose that X equals the time, measured in weeks, until a new pen runs out of ink. Based on past experience, Sam estimates that $\mathbb{E}(X) = 1$ week and $\mathbb{V}\text{ar}(X) = \frac{1}{4}$.

   To ensure that he has enough pens to last for 30 weeks at university, Sam decides to buy n pens at the start of the academic year. Use the Central Limit Theorem to determine how large n must be to ensure that the probability that Sam does not run out of pens while he is at university is at least 0.95.

8. A digital signal consists of a sequence of 0's and 1's. When the sequence is transmitted there is a chance that some of the digits in the sequence will be incorrectly transmitted (e.g. a '0' is transmitted when a '1' should be transmitted). To model this phenomenon, we will suppose that any digit in a sequence, independently of all other digits in the sequence, is incorrectly transmitted with probability 0.001.

   Now suppose that a signal consisting of 2000 digits is transmitted. Let X equal the number of errors in the transmission. Use the Poisson approximation to the Binomial distribution to *approximate* $\mathbb{P}(X > 2)$, the probability that there are more than 2 errors in the transmission.

9. Suppose that $X \sim \text{Poi}(\lambda)$ and $Y \sim \text{Poi}(\mu)$, and that X and Y are independent. Show, by computing $\mathbb{P}(X + Y = k)$ for $k = 0, 1, 2, ...$, that $X + Y \sim \text{Poi}(\lambda + \mu)$.

   [*Hint : You will need to make use of the Binomial theorem. Alternatively, you can make use of the fact that the probability mass function of a Binomial random variable sums to 1.*]

Exam Style Question

Jennie has a backlog of 80 emails to answer. For $i = 1, 2, \ldots, 80$, denote by X_i the time taken, in minutes, to answer the i-th email. Based on past experience, Jennie estimates that the time to answer each email, in minutes, has mean $\mathbb{E}[X_i] = 2$ and variance $\mathbb{V}\text{ar}(X_i) = 2$. Let $S = X_1 + X_2 + \ldots + X_{80}$ denote the total time taken, in minutes, to answer all of the emails.

(a) State the Central Limit Theorem. [1 mark]

(b) Stating carefully any assumptions that you make, use the Central Limit Theorem to approximate the probability that, starting at 8am, Jennie will have answered all of her emails by 11am. [6 marks]

(c) How much time should Jennie allocate to answering the emails to ensure that the probability that she will have answered all of the emails within the allocated time is at least 0.99. [5 marks]